BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 11231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** 11231

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The coordination number of each ion in NaCl crystal lattice is .....

(A) 2
(B) 4
(C) 6
(D) 8

Q2. The geometry of a molecule with two bond-pairs and one lone-pair of electrons around the central atom is .....

(A) linear
(B) angular
(C) pyramidal
(D) tetrahedral

Q3. Number of orbitals in M-shell (n=3) of an atom is .....

(A) 16
(B) 9
(C) 6
(D) 4

Q4. When moles of reactants and products are equal then equilibrium constant has units .....

(A) no unit
(B) mol.L-1
(C) L.mol-1
(D) L2.mol-2

Q5. Above -273°C, evaporation can take place at .....

(A) 0°C
(B) low temperature
(C) high temperature
(D) all temperatures

Q6. Mass of a sample of MgCl2 that contains 6.02×1023 Mg++ ions is .....

(A) 24g
(B) 35.5g
(C) 71g
(D) 95g

Q7. When energy of radiation A>C>B, then decreasing order of their wavelengths is .....

(A) A>B>C
(B) C>B>A
(C) B>C>A
(D) A>C>B

Q8. The number of orientations for an energy level n=2, l=1 is .....

(A) one
(B) three
(C) five
(D) seven

Q9. Deviation of gases from ideal behaviour depends upon .....

(A) temperature
(B) pressure
(C) nature of gas
(D) all of these

Q10. A specie that can act as a Lewis base, is .....

(A) SO2
(B) BF3
(C) ZnCl2
(D) CH2=CH2

Q11. Activation energy is the minimum energy required to cause ..... change.

(A) phase
(B) chemical
(C) temperature
(D) volume

Q12. Potential of standard hydrogen electrode is arbitrarily fixed as .....

(A) zero
(B) unity
(C) fractional
(D) negative

Q13. Ionic crystals are .....

(A) brittle
(B) soft
(C) volatile
(D) fusible readily

Q14. Energy required to cause a unit increase in the surface area of a liquid is .....

(A) vapour pressure
(B) viscosity
(C) surface tension
(D) isotropy

Q15. About 99% of universe is made up of .....

(A) gases
(B) liquids
(C) solids
(D) plasma

Q16. An expression that relates the rate of a reaction to the concentrations of reactants is .....

(A) Raoult's Law
(B) Henry's Law
(C) Rate Law
(D) Rate Constant

Q17. Oxidation state of chlorine in NaClO3 is .....

(A) -1
(B) -2
(C) +5
(D) +7

Q18. No work is done at constant .....

(A) temperature
(B) pressure
(C) mass
(D) volume

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40 | **Paper Code:** 11231

Q1. Attempt any ten of the following short questions. Each question carries 4 marks.

(i) What is Plank's Quantum Theory? Write down its postulates.

(ii) State Joule Thomson effect.

(iii) Why are crystalline solids anisotropic in nature?

(iv) What are the five series of spectral lines, observed for hydrogen atom?

(v) What is meant by resonance? Write down the basic rules for writing resonating structures.

(vi) Define plasma. How it occurs in nature?

(vii) Calculate the pH and pOH of 0.0005M H2SO4 solution.

(viii) What are dipole-dipole forces? Give examples.

(ix) Calculate the mole fraction of H2SO4 in one molal aqueous solution of H2SO4.

(x) How does Kc value help to predict the extent of a reversible reaction?

(xi) What are Lewis acids and bases? Give examples.

(xii) Calculate oxidation state of carbon in CH4, CH2Cl2, CHCl3 and CCl4.

(xiii) What are line emission and line absorption spectra?

SECTION-C

Maximum Marks: 27

Note: Attempt any three of the following questions. Each question carries 9 marks.

Q2.

(i) State and explain n+l rule.

(ii) What are limiting and excess reactants? How can limiting reactants be identified in a chemical reaction?

Q3.

(i) Explain definite shapes and sharp melting points of crystalline solids.

(ii) State and explain Le-Chatelier's principle.

Q4.

(i) State and explain Hess's law of constant heat summation.

(ii) Calculate the rate of diffusion of oxygen gas through a porous plate at 0°C, when rate of diffusion of hydrogen gas through the same plate at 0°C is 500cm/min.

Q5.

(i) Balance the following equations by ion-electron method: (i) $H_2O_2 + MnO_4^{-1} + H^{+1} \rightarrow Mn^{+2} + O_2 + H_2O$ (acidic medium) (ii) $I^{-1} + NO_2 \rightarrow I_2 + NO$ (acidic medium)

(ii) Give an account of London Forces in liquids.